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Proximity Effects of Wings on System Performance in a Multi-Wing Configuration

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The emergence of compactness requirements in modern Unmanned Aerial Vehicles (UAVs) and Private Air Vehicles (PAVs) has led to the development of new aircraft design concepts, such as Distributed Lift where multiple small span wings are used as a substitute to the conventional large span monowing. Wing-wing interactions were experimentally and numerically quantified as a function of gap and stagger across a wide range of different relative angles of attack (décalage) for two, three, and four-wing configurations. All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) using Clark-Y AR 2 semi-span wings. Numerical investigations were used to validate experiments and explore further configurations with increased numbers of wings. The proximity effects and the zone of influence at different gap and stagger locations were systematically characterized by measuring the changes in aerodynamic force coefficients of the combined wing system. A wide range of beneficial and detrimental combinations of gap, stagger, and décalage were determined for optimal placement of wings in a distributed lift system. The effect of adding wings to a base configuration was explored through aerodynamic forces, surface pressure contours, and induced angle of attack changes.

I. Nomenclature

St	=	Stagger (Lateral Spacing)
G	=	Gap (Longitudinal Spacing)
$\propto$	=	Geometric Angle of Attack
$\propto_{L=0}$	=	Zero Lift Angle of Attack
α_i	=	Induced Angle of Attack
c	=	Chord
b	=	Wingspan
AR	=	Aspect Ratio; $AR = \frac{b^2}{S}$
U_∞	=	Freestream Velocity
C_L	=	Lift Coefficient; $C_L = \frac{Lift}{(1/2)*\rho*V^2*S}$
C_{L_α}	=	Lift Curve Slope
C_D	=	Drag Coefficient; $C_D = \frac{Drag}{(1/2)*\rho*V^2*S}$
L/D	=	Lift to Drag Ratio

Subscripts

Sweep	=	Sweeping Wing
Neighbor	=	Neighbor Wing
Biplane	=	Two-Wing Biplane Group
Tandem	=	Two-Wing Tandem Group
sys	=	All Wings Combined System

II. Introduction

Unmanned aerial vehicles (UAVs), drones, and air taxis have become increasingly popular for various commercial, entertainment, and military applications due to their small footprint and niche capabilities. People/companies such as

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Distribution Statement A: Approved for Public Release; Distribution is Unlimited. PA# AFRL-2023-3435 d'Ecquevilly [1], Faradair [2], SE Aeronautics [3], etc. have unveiled designs in Fig. 1 with multiple lift-producing surfaces and rotors working in close proximity to one another. In 1907, Marquis d'Ecquevilly, a French engineer, aimed to test the theory of whether adding more wings would allow an aircraft to generate more lift. He created the d'Ecquevilly Multiplane which had six pairs of wings arranged within an elliptical frame with the pilot and engine in the center. The Faradair BEHA is a hybrid-electric triple box wing aircraft designed by a UK aircraft manufacturer to carry around 18 passengers and be the sustainable future of regional air mobility. SE Aeronautics aims to change the airliner industry with their SE200 design that is stated to be a more efficient aircraft with quicker manufacturing time due to their unique approach of three thin wings.

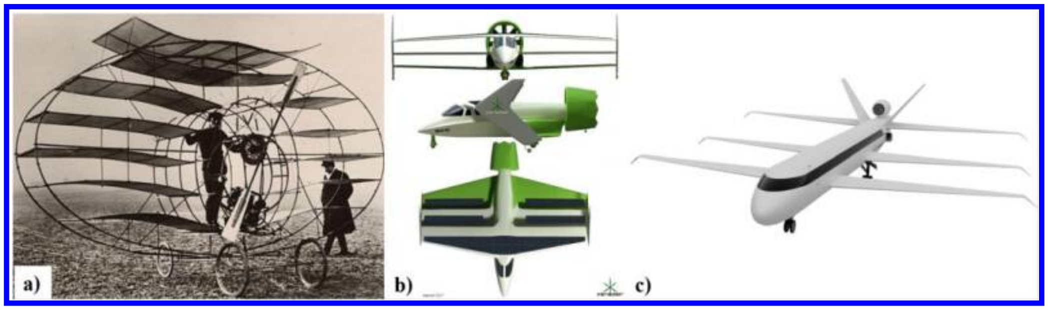

Fig. 1 Multiple Wing Concepts of a) d'Ecquevilly Multiplane [1], b) Faradair BEHA [2], c) SE Aeronautics SE200 [3]

Building upon this idea of multiple wing designs, distributed lift, as seen in Fig.2, aims to minimize the wingspan and increase "compactness" by distributing the total wing area of a conventional monowing to multiple small span wings. Hypothesized additional benefits of such a design are listed in a preliminary studies paper by Truszkowski and Altman [4], some of which were increased maneuverability, lighter weight, and higher damage tolerance.

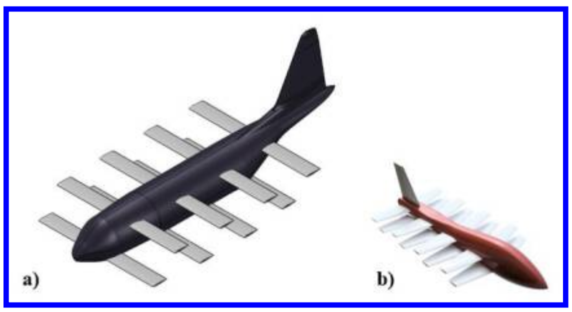

Fig. 2 OpenVSP Distributed Lift (Multi-Wing) Aircraft Concepts of a) Cargo inspired aircraft and b) UAV

Biplanes were revisited through experimental investigations by Kang et al. [5,6] on the effects of gap and stagger on biplanes with endplates. In a biplane configuration, the gap and stagger had a significant influence on performance. Using Particle Image Velocimetry (PIV), it was observed that decreasing the gap between the two wings resulted in a smaller downwash angle off the trailing edge, which in turn resulted in a lower lift. PIV also disproved the hypothesis of a jet of increased velocity being created between the two wings due to a reduction in area both geometrically and aerodynamically because of larger downwash angles produced by the upper wing. The study observed that beyond a 1-chord gap and stagger, the benefits of further wing-wing separation rapidly diminished with minimal difference in lift coefficient beyond that point.

Initial experimental investigations on distributed lift were done earlier by Mongin et al. [7,8,9] where several multi-wing configurations were tested at the University of Dayton Low Speed Wind Tunnel and at the Vertical Wind Tunnel at the U.S. Air Force Research Laboratory. The experimental results were also compared with the numerical investigations from Vortex Lattice Method codes. In these investigations, all the wings in a given multi-wing configuration were placed at fixed spacings and equal geometric angles of incidence. The results showed that the multi-wing configurations were able to achieve around 50% of the aerodynamic efficiency of that of a monowing, even approaching 61% in a multi-wing configuration. With the feasibility of this concept confirmed, full span models of a multi-wing generic transport platform with three different wing sets were tested in an angle of attack and sideslip angle sweep at the Air Force Research Laboratory Vertical Wind Tunnel (AFRL-VWT). The results showed sideslip and Reynolds number insensitivity for lift. Although gap, stagger, and offset were not held constant between the different configurations, there was a general trend of a decreasing lift curve slope as the number of wings increased.

A more fundamental approach was undertaken by Jestus et al. [10] to better optimize the distributed lift configurations by quantifying the isolated wing-wing interactions between two wings in close proximity. The study confirmed that placing a wing directly upstream of another wing in close proximity of 1 chord to 0.5 chord stagger can lead to a decrease in aerodynamic efficiency of the aft wing. A secondary wing along a diagonal of high gap and low forward stagger saw the least amount of performance decrement as determined by the mean system L/D contour of Fig. 3. Slight asymmetry is also visible, emphasizing the importance of stagger and gap direction. System analysis of the two wings combined saw the classical biplane configuration with 0c stagger and 1c gap to have high performance at several angle of attack combinations.

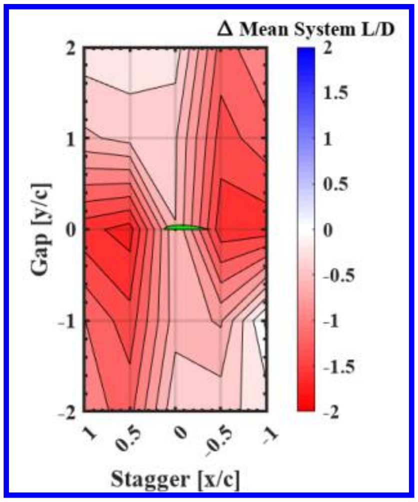

Fig. 3 Difference in Mean System L/D from Mean Ideal System Efficiency across all angles of attack at various gap and stagger locations representing the Zone of Influence [10]

Extending upon the previous study, Jestus et al. [11] observed the interactions among three identical wings as a function of proximity. Analyzing the changes in performance of each individual wing can allow inter-wing optimization for more complex multi-wing concepts. A two-wing group in either a true biplane (0c St, 1c G) or true tandem (1c St, 0c G) was moved around a wing sweeping through angle of attack to study the two extreme two-wing configurations. Figure 4 displays the mean lift performance within the biplane/tandem two-wing group subtracted by the baseline of a biplane or tandem interaction. The dots represent the centroid between the two wings pair where for the biplane group, there will be a wing above and below the dot. For the tandem, there will be a wing in front and behind the dot. Although wings in a biplane configuration performed the best, they were much more sensitive to the addition of another wing than the tandem configurations as indicated by the darker red contour color.

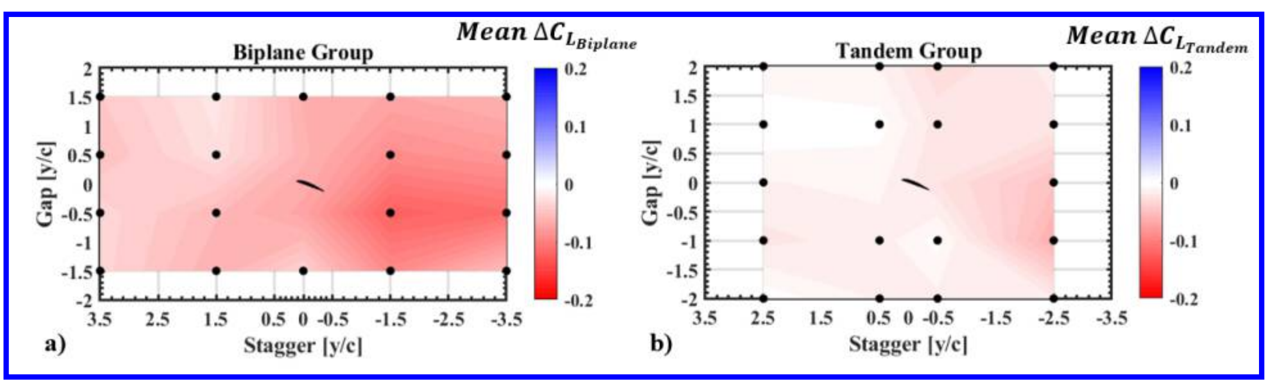

Fig. 4 Difference in Mean C_L in a) Biplane Configurations or b) Tandem Configurations from Biplane/Tandem Baseline across all angles of attack at various gap and stagger locations [11]

As distributed lift systems are explored in more depth, the number of wings used, the size of the individual wings, and their placement should be optimized. These experiments were performed with the objective of inter-wing optimization focused on the system analysis of all wings combined as the number of wings increases up to four wings.

III. Experimental Setup

All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) in the open-jet configuration. The tunnel has a 16:1 contraction ratio and a test section size of 76.2 cm by 76.2 cm by 137 cm with velocities ranging from 6.7 m/s to 36.7 m/s. The maximum turbulence intensity was measured to be 0.1% at 15 m/s using a hot-wire anemometer.

A. Experimental Setup

The experiments were conducted with several identical Clark Y AR 2 semispan wings with a chord of 0.127 m and span of 0.254 m. These wings were placed at various stagger and gap locations as well as relative angles of attack (décalage). The wing configurations included two, three, and four wings placed at several gap and stagger locations around one another. The two-wing configurational study was discussed in detail in Jestus et al. [10] and the three-wing preliminary results are discussed in Jestus et al. [11]. The four-wing study was conducted as a part of this investigation and was compared against system level performance of a two-wing and a three-wing study. Two approaches were used to study the four-wing configurations. In the first method, two biplane configurations with 1c-gap distance were placed at different locations around each other. In the second method, two tandem configurations with 1c-stagger distance were placed around each other.

Table 1 shows a summary of the full experimental test matrix. Neighbor Wings were positioned at these gap and stagger locations at a fixed angle of attack while the Sweeping Wing angles were changed from -20° to 20°. Then the Neighbor Wing angle was changed at the same gap and stagger location and the Sweeping Wing angle of attack sweep is performed again. The same procedure was used at different gap and stagger locations. The five Neighbor Wing angles of attack were selected to capture post negative stall, zero lift angle ($\propto_{L=0}$), two intermediate attached flow angles, and then the point of maximum lift ($C_{L_{max}}$) based on a single Clark Y AR 2 wing. When introducing the three and four wing tests, the centroid of the two Neighbor Wings in their respective biplane or tandem group was positioned at these stagger and gap locations at a fixed angle of attack while the Sweeping Wing angles (α_{Sweep}) were changed from -20° to 20°. Then the biplane or tandem's angle was changed at the same stagger and gap location before a Sweeping Wing angle of attack sweep is performed again. The baseline individual wing Reynolds number tested was 250,400 at a freestream speed of 30 m/s.

Table 1 Experimental Test Matrix for Two, Three, and Four Wing

Two-Wing (Re= 250,400)				
Two Wing Group	Stagger Distances [x/c]	Gap Distances [y/c]	Sweeping Wing Alpha Range [°]	Neighbor Wing Alpha Range [°]
----------------	1, 0.5, 0, -0.5, -1	2, 1, 0, -1, -2	-20 to 20	-10, -3, 8, 13, 20
Three-Wing (Re= 250,400)				
Two Wing Group	Stagger Distances [x/c]	Gap Distances [y/c]	Sweeping Wing Alpha Range [°]	Neighbor Wing Alpha Range [°]
(0c St, 1c G) Biplane	3.5, 1.5, 0, -1.5, -3.5	1.5, 0.5, -0.5, -1.5	-20 to 20	-3, 3, 8
(1c St, 0 G) Tandem	2.5, 0.5, -0.5, -2.5	2, 1, 0, -1, -2	-20 to 20	-3, 3, 8
Four-Wing (Re= 250,400)				
Two Wing Group	Stagger Distances [x/c]	Gap Distances [y/c]	Sweeping Wing Alpha Range [°]	Neighbor Wing Alpha Range [°]
(0c St, 1c G) Biplane	4, 2, -2, -4	1, 0, -1	-20 to 20	-3, 3, 8
(1c St, 0 G) Tandem	2, 0, -2	2, 1, -1, -2	-20 to 20	-3, 3, 8

The nature of this study necessitates a slightly adjusted reference frame from biplane and previous distributed lift work. In the two-wing and three-wing cases, the coordinate position is from the biplane centroid to the Sweeping Wing leading edge. In the four-wing case, the coordinate position is from the centroid of both wing pairs. The axis reference convention is that the x-coordinate represents the stagger distance in chord lengths between the wings and the y-coordinate is the gap distance as annotated in Fig. 5. Figure 6 displays all of the four-wing biplane configurations experimental tested as a visual reference where the red wing is the Sweeping Wing and green is the Neighbor Wings. Figure 7 displays the four-wing tandem configurations with red once again as the Sweeping, but the Neighbor Wings now depicted in blue to highlight the tandem orientation.

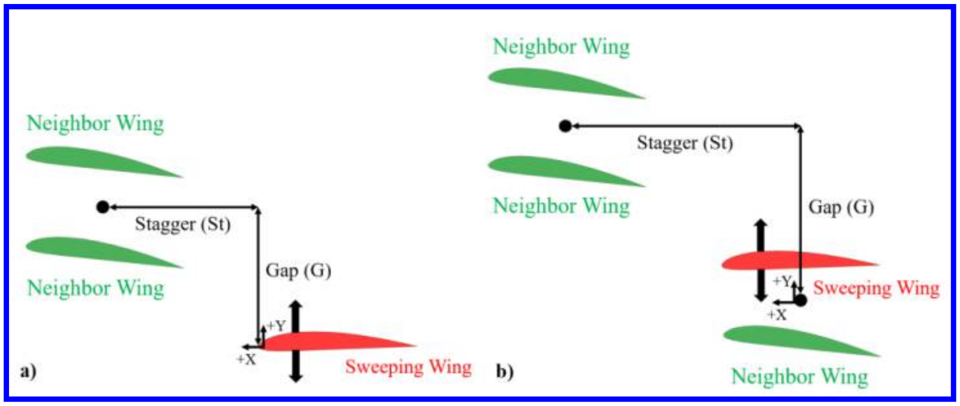

Fig. 5 Nomenclature difference for a) Three-Wing configurations and b) Four-Wing configurations

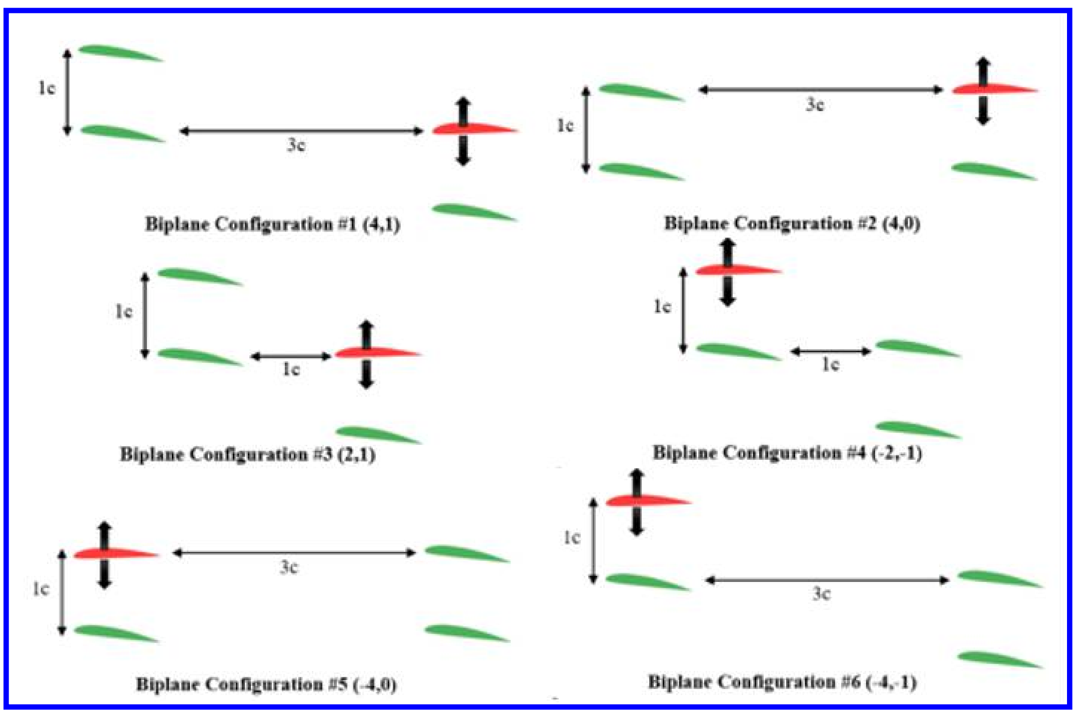

Fig. 6 Experimentally tested Four-Wing *biplane* configurations

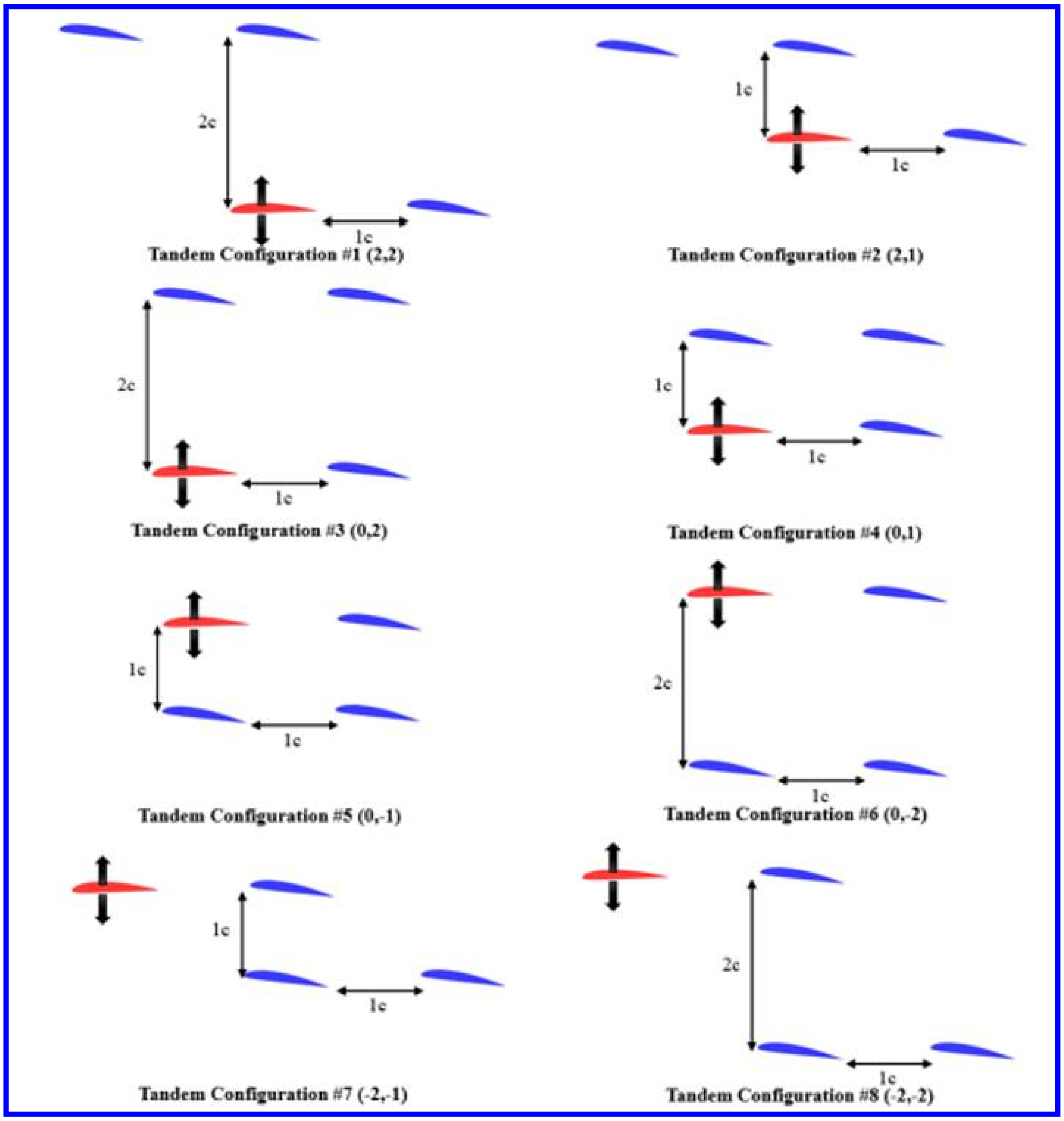

Fig. 7 Experimentally tested Four-Wing *tandem* configurations

The Sweeping Wing was attached to a 6 component Analog ATI Gamma Net F/T sensor [12] which was mounted on top of a PDV PT-GD201 rotary stage [13] to adjust the angle of attack. Similarly, a Neighbor Wing was connected to a 6 component Digital ATI Gamma Net F/T sensor [12] on top of a PDV PT-GD201 rotary stage [13]. The second Neighbor Wing measured the forces and torques with a 6 component ATI Mini40-E [14] and another PDV PT-GD201 rotary stage [13] to rotate angle of attack. The last Neighbor Wing also used an ATI Mini40-E [14] and another PDV PT-GD201 rotary stage [13] to measure forces and torques with angle of attack rotation. Table 2 below summarizes the type of force balance and the rotary stage used for each of the four wings. Figure 8 shows a schematic of the experimental setup in the UD-LSWT.

Table 2 Experimental Sensors and Rotary Stages

Wing	Force Balance	Force Balance Calibration	Rotary Stage
Wing 1 (Sweeping)	Analog ATI Gamma Net F/T	SI-65-5	PDV PT-GD201
Wing 2 (Neighbor)	Digital ATI Gamma Net F/T	SI-65-5	PDV PT-GD201
Wing 3 (Neighbor)	ATI Mini40-E	SI-40-2	PDV PT-GD201
Wing 4 (Neighbor)	ATI Mini40-E	SI-40-2	PDV PT-GD201

The ATI Gamma Net F/T sensor was oriented with the normal and axial forces measured through the X and Y axis. The sensing ranges of the Gamma in the SI-65-5 Calibration of X and Y is 65 N with an uncertainty of 0.75% and 1.00% respectively along with the torque range of 5 Nm and uncertainty of between 1.00% to 1.50% depending on the axis. The ATI Mini40-E Net F/T sensor was oriented with the normal and axial forces measured through the X and Y axis. The sensing ranges of the Mini40 in the SI-40-2 Calibration of X and Y is 40 N with an uncertainty of 1.25% for both axes along with the torque range of 2 Nm and uncertainty of 1.25% to 1.50%. A sampling rate of 1000 Hz was used during data acquisition. The PDV GD-201 rotary stage instrument uncertainty in angle of attack was 0.01°. In order to find the 0° angle of attack for each individual wing, the zero lift angle ($\propto_{L=0}$) was found through force measurement for each wing independently without the presence of any other wings in close proximity.

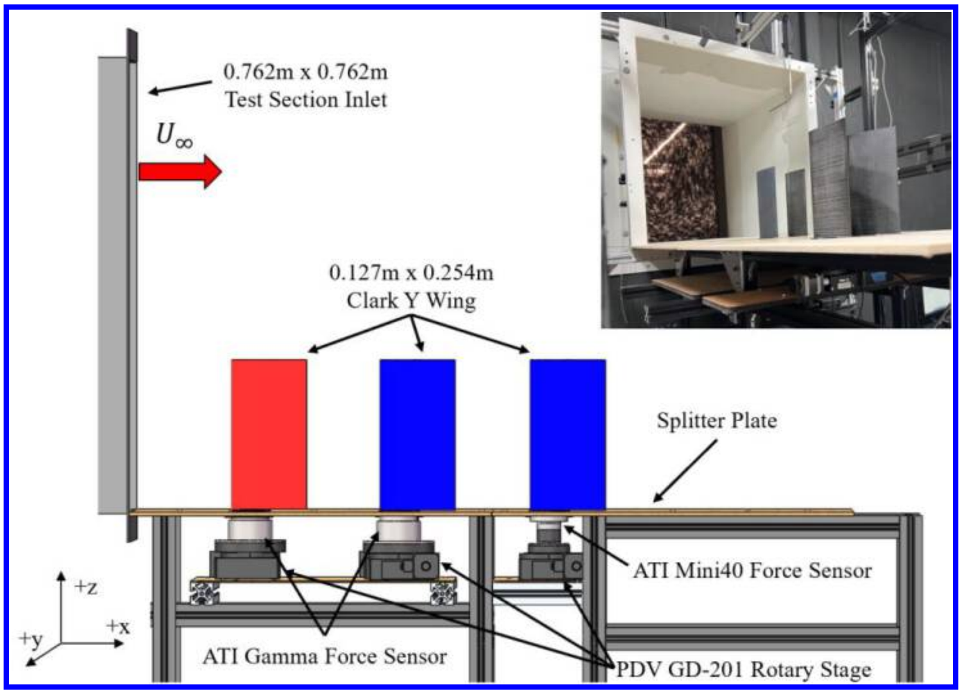

Fig. 8 CAD schematic of experimental setup in University of Dayton Low-Speed Wind Tunnel along with a photograph in top right corner

B. Numerical Setup with FlightStream®

FlightStream® [15] is a potential flow solver capable of two main solver types: pressure-based and vorticity. This study utilized the pressure-based solver for its closer agreement with prior UD-LSWT experimental results [16]. The solver determines the pressure fields on the model geometry to calculate the aerodynamic loads. The combination of source and sink panels along with doublet panels on the trailing edges of the wings provides results for the pressure-based potential flow solver.

Figure 9 represents the surface mesh for the simulations with one of the four-wing configurations as an example. Each individual wing has a structured mesh with 60 node points chordwise and 161 node points spanwise, generated in an OpenVSP [17] model and imported to FlightStream as a Pittsburgh Three Dimensional (.p3d) file. A dual side growth rate of 1.08 was applied to resolve the chordwise mesh on the LE and TE. A splitter plate was added with the same dimensions as the one used in the experimental study, measuring 0.9144 m by 1.524 m, to closely match all conditions. FlightStream generated a "sheet" to represent the splitter plate with a mesh of 40 node points by 80 node points. A trailing edge line was marked, and a wake termination node placed on the node connecting the TE and splitter plate for each wing. These termination nodes are required to prevent any wake strands that flow along the geometry making it coincident with the boundaries of the fluid and therefore a violation of Helmholtz's second theorem.

The relevant solver parameters that were chosen for all simulations presented are summarized in Table 3 for reference. The atmospheric properties were matched with the average values recorded during testing in the UD-LSWT.

In the solver initialization, the wake is calculated 5 meters downstream of the wings with surface proximity avoidance enabled. The steady solver was run until convergence tolerance was met for the velocity and pressure residuals. A transitional turbulent viscous boundary layer was simulated with viscous coupling enabled. The axial model was used for flow separation.

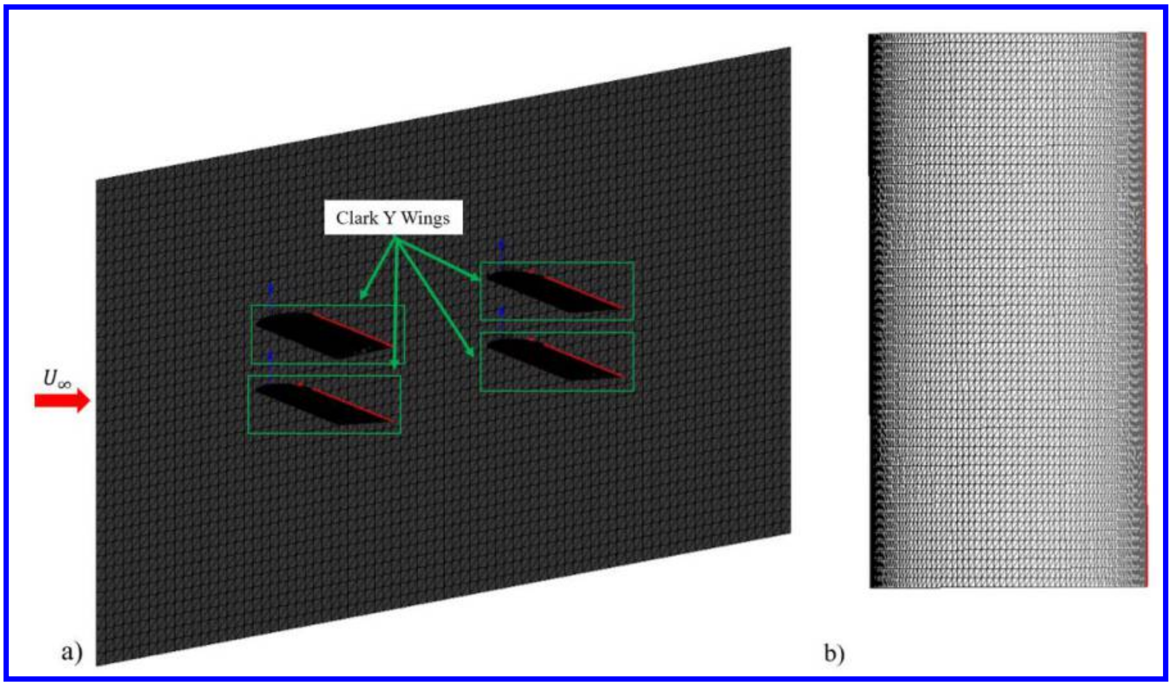

Fig. 9 FlightStream surface mesh of a) four wings and splitter plate isometric view and b) top view of an isolated wing

Table 3 Steady Solver Parameters

Parameter	Setting
Viscous Boundary Layer	Transitional Turbulent
Viscous Coupling	Enabled
Flow Separation	Axial Model
Convergence Threshold	1.00 10^{-5}

IV. Results

This section will establish critical baseline definitions that are used for analysis of these multi-wing configurations. Once the baselines are understood, analysis will be presented in two perspectives, proximity effects for varying number of wings and then the effect of adding wings on select base configurations.

A. Baselines

To attain a comparable single wing baseline case, one of the Clark Y wings was tested separately at different positions in the spanwise direction and downstream of the wind tunnel inlet to ensure repeatability at various spatial locations in the test section. Their associated different force balances were also used to ensure repeatability across different sensors and calibrations. Figure 10 shows the averaged lift curve of the single wing from the different locations where the dashed lines represent the error bars. The baseline cases see negative stall around 10° and positive stall around 18° with a $C_{L_{Max}}$ of 1.1. The lift curve slope ($C_{L_\propto}$) in the linear region of the averaged curve is calculated to be 0.057 deg^{-1} which is around $C_{L_\propto} = \pi\alpha$. This is consistent with low AR wing behavior. Using Helmbold's low aspect ratio equation [18] in Eq. (1), the aspect ratio corrected lift curve slope of the averaged baseline is calculated with C_{l_α} as 2π for thin airfoils and an effective AR of 2.8. The corrected lift curve slope was derived to be 0.0564 deg^{-1} and is also plotted in Fig. 8 for comparison.

$$C_{L_\alpha} = C_{l_\alpha} \frac{AR}{\left(\frac{C_{l_\alpha}}{\pi}\right) + \sqrt{\left(\frac{C_{l_\alpha}}{\pi}\right)^2 + AR^2}} \tag{1}$$

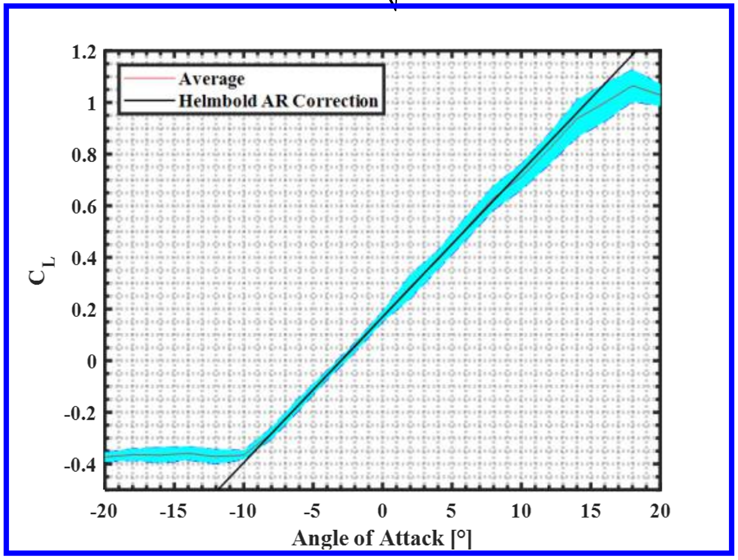

Fig. 10 Baseline AR 2 Clark Y Single-Wing Lift Curve

When referring to a two-wing biplane or tandem group, a baseline is desired which accounts for the existing wing-wing interactions to assess the incremental improvement or decrement in performance of the ensemble as a function of the addition and proximity of wings. From the two-wing study performed in Jestus et al. [10], there are experimental data for two wings in these orientations at the same angles of attack of -10°, -3°, 8°, 13°, 20°. Linear interpolation can then be performed to obtain the C_L of the biplane or tandem as a single system at 3° angle of attack assuming a continuous linear range. Figure 11 shows this linear trend for both the system C_L of the biplane and tandem configurations. Both configurations display a deviation in the lift curve slope from the standalone wing, with the tandem exhibiting behavior similar to an AR reduction compared to the biplane.

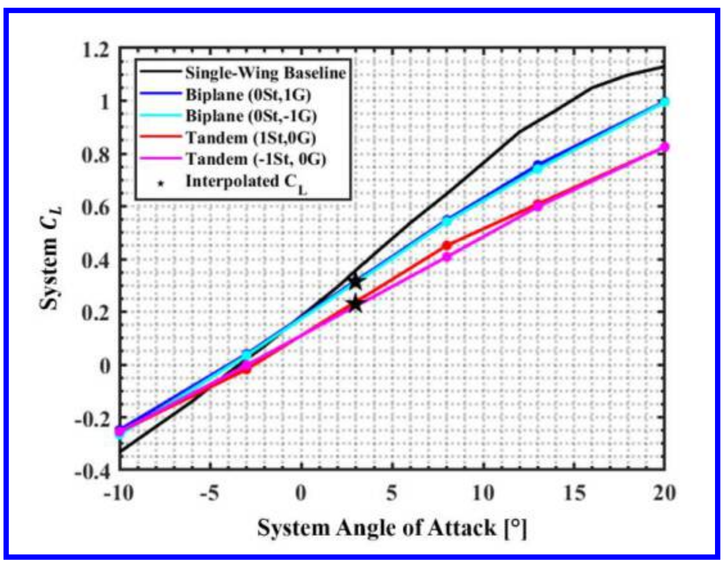

Fig. 11 Two-Wing Biplane and Tandem combined lift coefficient compared to Single-Wing

The system lift to drag ratio (L/D) was selected as the metric to quantify the overall wing-wing interactions. The system L/D for a specific configuration was calculated using the measured forces from all wings added together as seen in Eq. (2). An Ideal System Efficiency (ISE) as defined by Eq. (3) is provided to compare the extent of aerodynamic interactions when compared to the linear superposition of the lift and the drag forces. The Single-Wing Baseline is used to extract the “ideal” lift and drag coefficient for each wing at their specific angle of attack. These determined lift and drag coefficients are then added and divided to get an ideal total L/D, without any positive or negative wing-wing interactions, known as the ISE. The example below is of a four-wing configuration. For instance, when the Sweeping Wing is at 14° and the three Neighbor Wings are at 8°, the ISE is calculated by taking the sum of the Single-Wing Baseline lift at 14° and thrice the Single-Wing Baseline lift at 8°, and then dividing it by the sum of the Single-Wing Baseline drag at 14° and thrice the Single-Wing Baseline drag at 8°.

$$L/D_{sys} = \frac{\sum_{i=1}^{n} L_n}{\sum_{i=1}^{n} D_n} \tag{2}$$

$$Ideal\ System\ Efficiency\ (ISE) = \frac{L_{Baseline\ Sweep} + n * L_{Baseline\ at\ \alpha_2}}{D_{Baseline\ Sweep} + n * D_{Baseline\ at\ \alpha_2}} \tag{3}$$

Efficiencies higher than the linearly superimposed ratio of the forces represents positive aerodynamic interference and vice-versa. Figure 12 displays the ISE for the two, three, and four-wing systems at three different α_{Group} to allow comparison between the effect of angle of attack and number of wings on a baseline. When the wings are at the $\propto_{L=0}$, there is a large difference between the two, three, and four wing systems, but as the fixed angle of attack increases, these differences become minimal. Despite an increased number of wings, the maximum L/D never surpasses the maximum of the one-wing baseline. Once α_{Group} rotates beyond 3°, the ISE becomes completely positive as the lift generated by the fixed wings negate any negative lift produced by the sweeping wing. Each two, three, and four-wing experimental configuration can have their system performance plotted along with their respective ISE baseline to not only determine which configurations perform the best but determine if any use positive proximity interactions between wings to surpass the ISE. Notable in the figure is that for positive angles of attack, the location of the maximum lift to drag ratio is independent of any interference effects related to the number of wings. Additionally, for positive group angles of attack, there is close correspondence with the baseline performance.

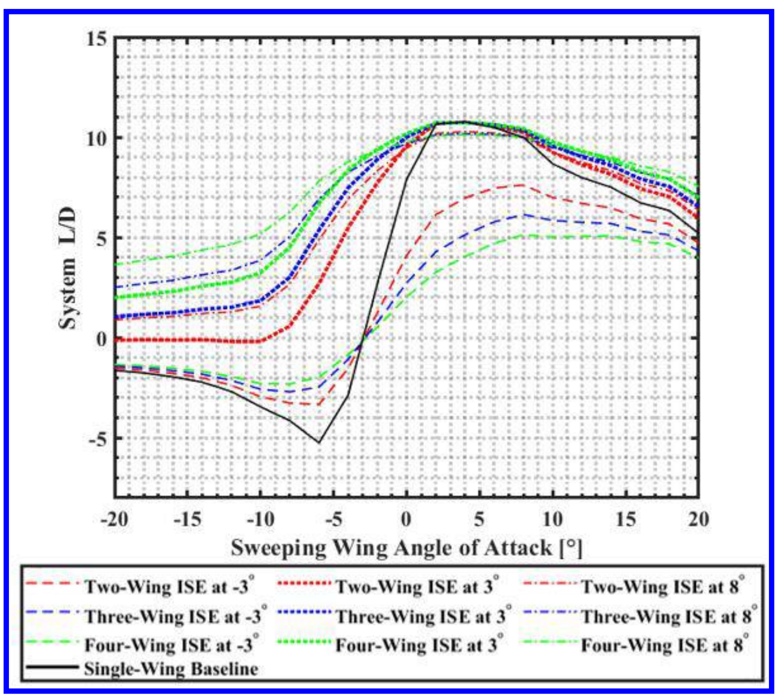

Fig. 12 Comparisons of Ideal Multi-Wing System Baselines, angles of attack refer to a single Sweeping Wing with all other wings fixed at the stated angle

B. Proximity Effects

Multiple different configurations were tested with different numbers of wings with systematically varying gap, stagger, and decalage adjustments. The results were analyzed in both an individual wing and system reference frame. Analysis of the two-wing and three-wing configurations are presented in previous papers by Jestus et al. [10,11]. Exhaustive analysis of the four-wing configurations can be found in the thesis by Jestus [16].

For the most part, the trends correspond with those observed in the fewer wing configurations. In terms of individual or two-wing group performance, upstream wings perform similarly to what is expected from their respective baselines. When these wings move downstream of other Neighbor Wings, the performance experiences disruptions as efficiency decreases. In a system analysis, as the number of wings increases, two main trends are observed: convergence of all configurations to a similar L/D curve and increasing deviation from the ideal system efficiency (ISE).

Figure 13 plots the system L/D of all 6 four-wing biplane configurations at the three different Neighbor Wing angle of attack sets. The Single-Wing Baseline and ideal system efficiency (ISE) are plotted along with the multi-wing configurations. When the three fixed wings are at an $\alpha_{Neighbor}$ of 3°, the configurations follow the ISE relatively closely with the high performing configurations maintaining this closeness through all Sweeping Wing angles. A transition to higher $\alpha_{Neighbor}$, begins a trend of separation from the ISE along with minimal differences between these distinct configurations.

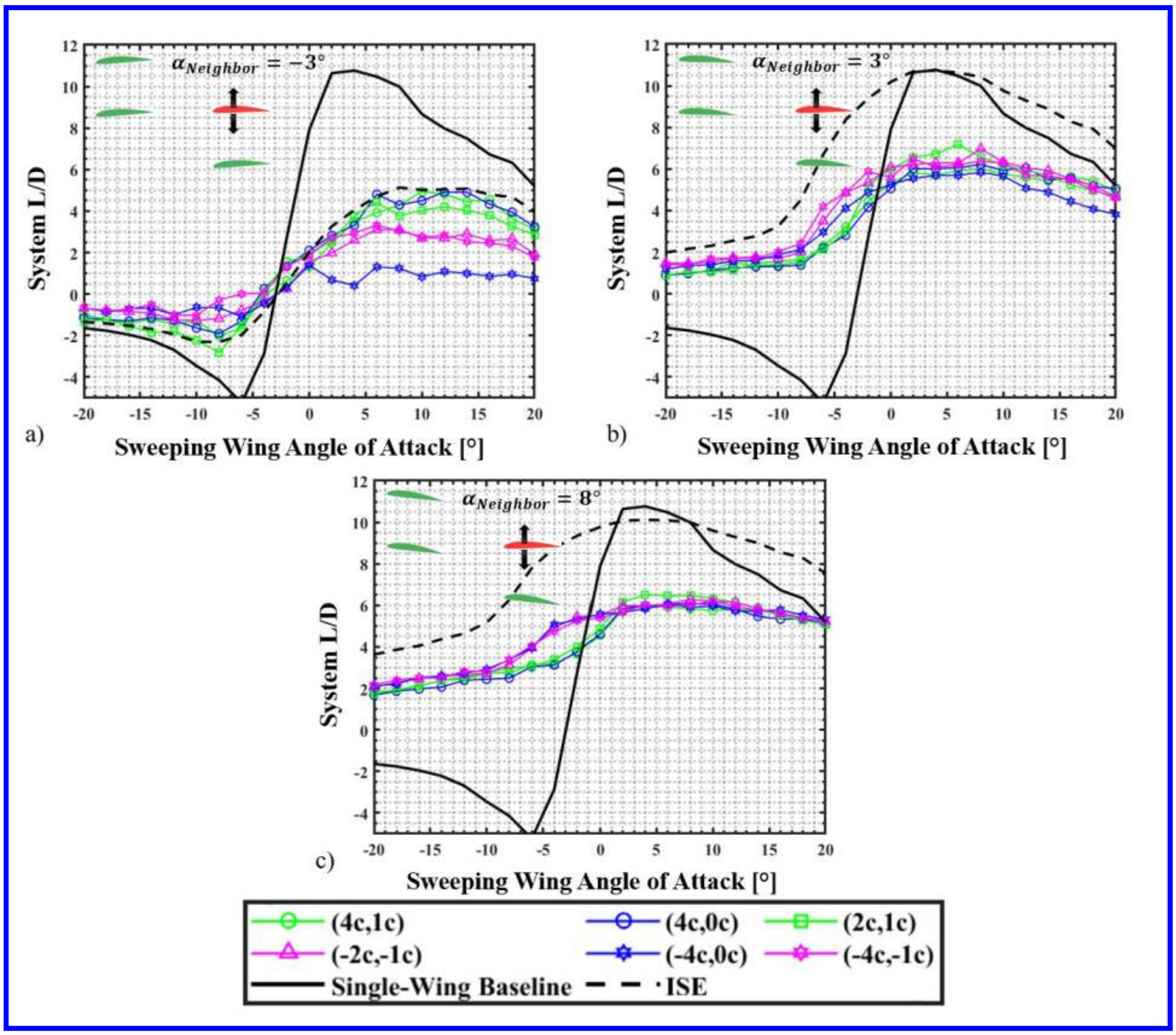

Fig. 13 System L/D of various four-wing *biplane* configurations as Sweeping Wing sweeps and Neighbor Wings are fixed at a) -3°, b) 3°, and c) 8°

Figure 14 is a side-by-side comparison of all the four-wing *tandem* configurations as the Neighbor Wing angle of attack changes. Very similar trends appear in the *tandem* configurations as seen in the *biplane* configurations where there is not much of a difference between the different orientations. The four-wing *tandem* configurations do experience a large variation from the ISE at -3° even from the high performing cases which were able to match the ISE in the four-wing *biplane* configurations. As $\alpha_{Neighbor}$ increases, the difference from the ISE continues. In the case of a 3° and 8° $\alpha_{Neighbor}$, the system L/D is relatively consistent across all Sweeping Wing angles of attack as seen by the flattened L/D curve. When the Neighbor Wing angle of attack is at 3°, there does seem to be more of the expected curvature for the L/D, with a more clearly defined maximum.

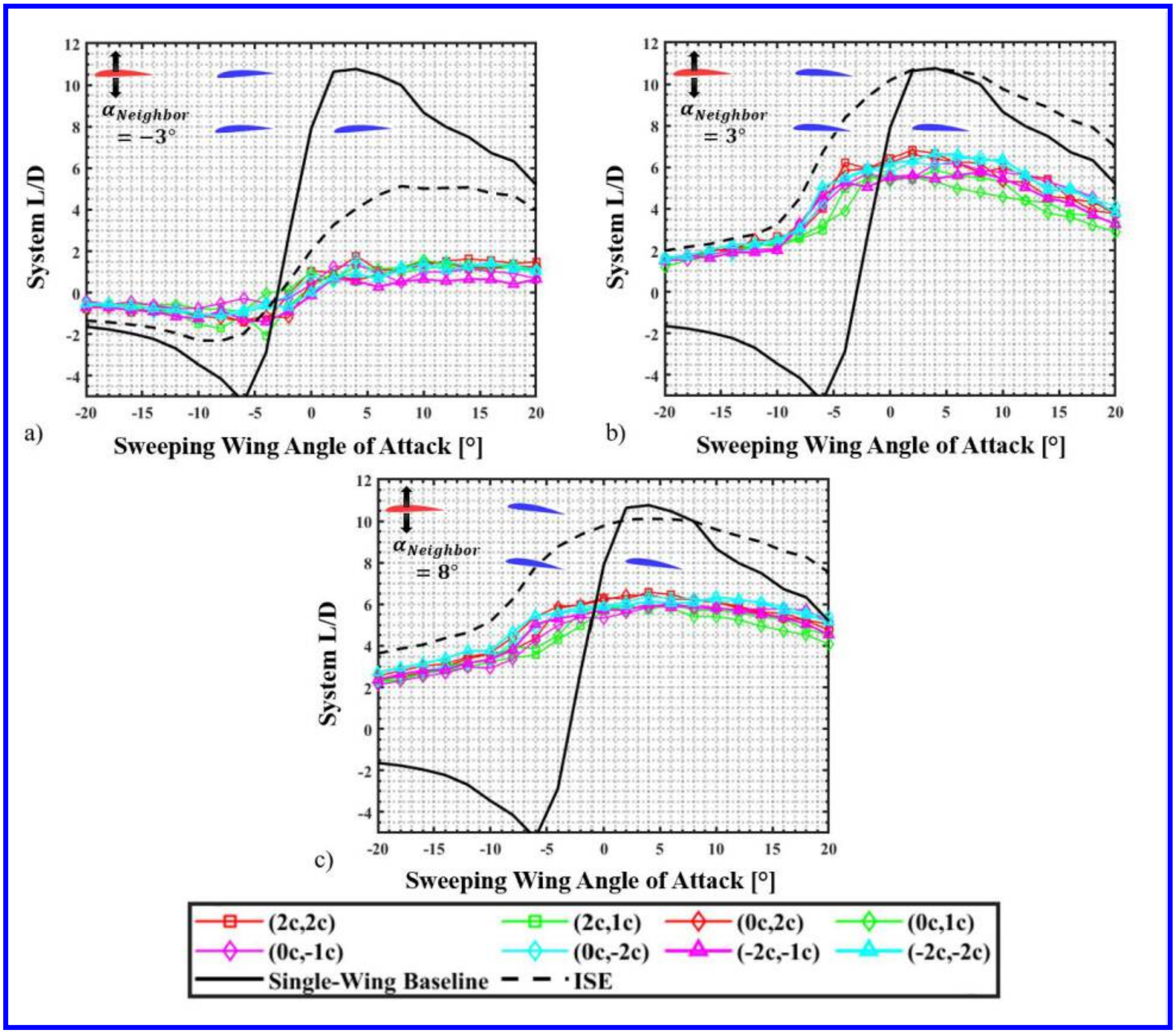

Fig. 14 System L/D of various four-wing *tandem* configurations as Sweeping Wing sweeps and Neighbor Wings are fixed at a) -3°, b) 3°, and c) 8°

The maximum L/D potential irrespective of the configurations for the different wing number sets at their respective $\alpha_{Neighbor}$ are visualized in Figure 15. These values are the maximum recorded L/D from each system analysis performed in their respective chapters. The baseline of a single wing achieves a maximum L/D of around 11 at 4°. Figure 13 serves as a reminder that the two-wing configurations were able to match the ISE very closely and achieve similar efficiency to that of a single standalone wing. As wings were generally added, the maximum L/D decreased to 6 which is about 60% of the monowing L/D, regardless of the wing placements relative to each other. The effect of different $\alpha_{Neighbor}$ was shown to have large implications during system analysis, but according to the heatmap may not have significantly impacted the maximum L/D of these configurations with only slight differences in L/D within ±2 across a wide range of $\alpha_{Neighbor}$ from -3° to 13°.

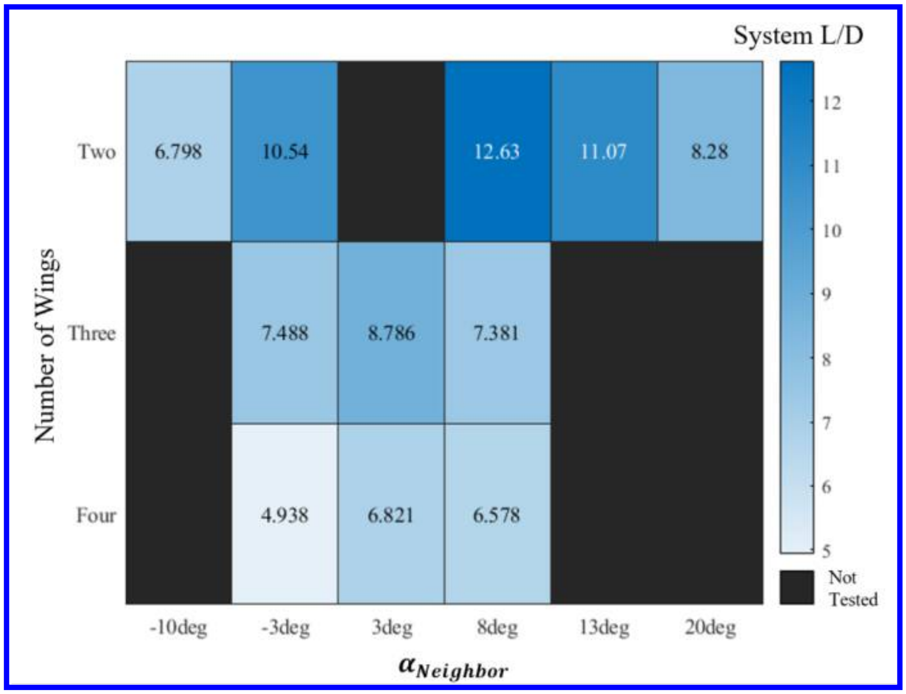

Fig. 15 Heatmap of highest measured system L/D throughout all two, three, and four wing studies

C. Effect of Adding Wings to a Vertical Stack, Tandem, and Diagonal Configuration

Understanding of this convergence in performance irrespective of configuration requires identification of an aerodynamic source for the behavior. To better isolate the effect of adding wings to a set configuration, three configurations are selected at constant gap, stagger, and décalage. The combination of a vertical stack, tandem, and diagonal isolate the effects on the performance of the combined multi-wing system of having only a gap component, only a stagger component, or a combination of both. For reference, these three base configurations are visually represented in Fig. 16 below.

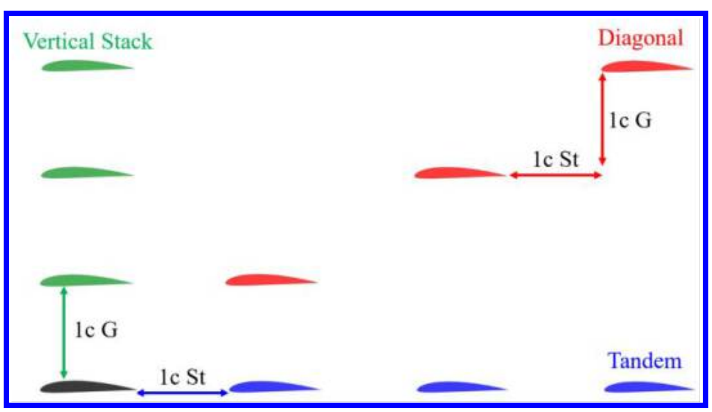

Fig. 16 Three extreme base configurations of wings arranged in a Vertical Stack, Tandem, and Diagonal

Figure 17 illustrates the effect on lift from adding wings in vertical stacked, tandem, and diagonal configurations where the color red represents a two-wing configuration, blue is three-wing, and green is four-wing. The addition of wings decreases the lift curve slope to varying degrees, contingent upon the configuration. The vertically stacked configuration demonstrates an overall insensitivity to the addition of wings when evaluated by FlightStream simulations. However, disparities in the magnitude of the lift curve slopes for two- and three-wing configurations are evident between experiments and FlightStream. While the number of wings may not significantly affect the lift curve slope in vertically stacked configurations, the addition of multiple wings substantially decreases the lift curve slope compared to a single-wing setup. In contrast, the tandem configuration shows a slight increase in sensitivity to the number of wings. The system lift coefficient tends to drop in magnitude at most angles when additional wings are incorporated. The diagonal configuration seems to strike a balance between vertically stacked and tandem configurations in both geometry and sensitivity to the number of wings. It is the configuration most apparently approaching stall or at least plateauing. According to FlightStream, the different configurations seem to approach a similar maximum lift coefficient, but there is significant disagreement with experimental results at these high angles of attack.

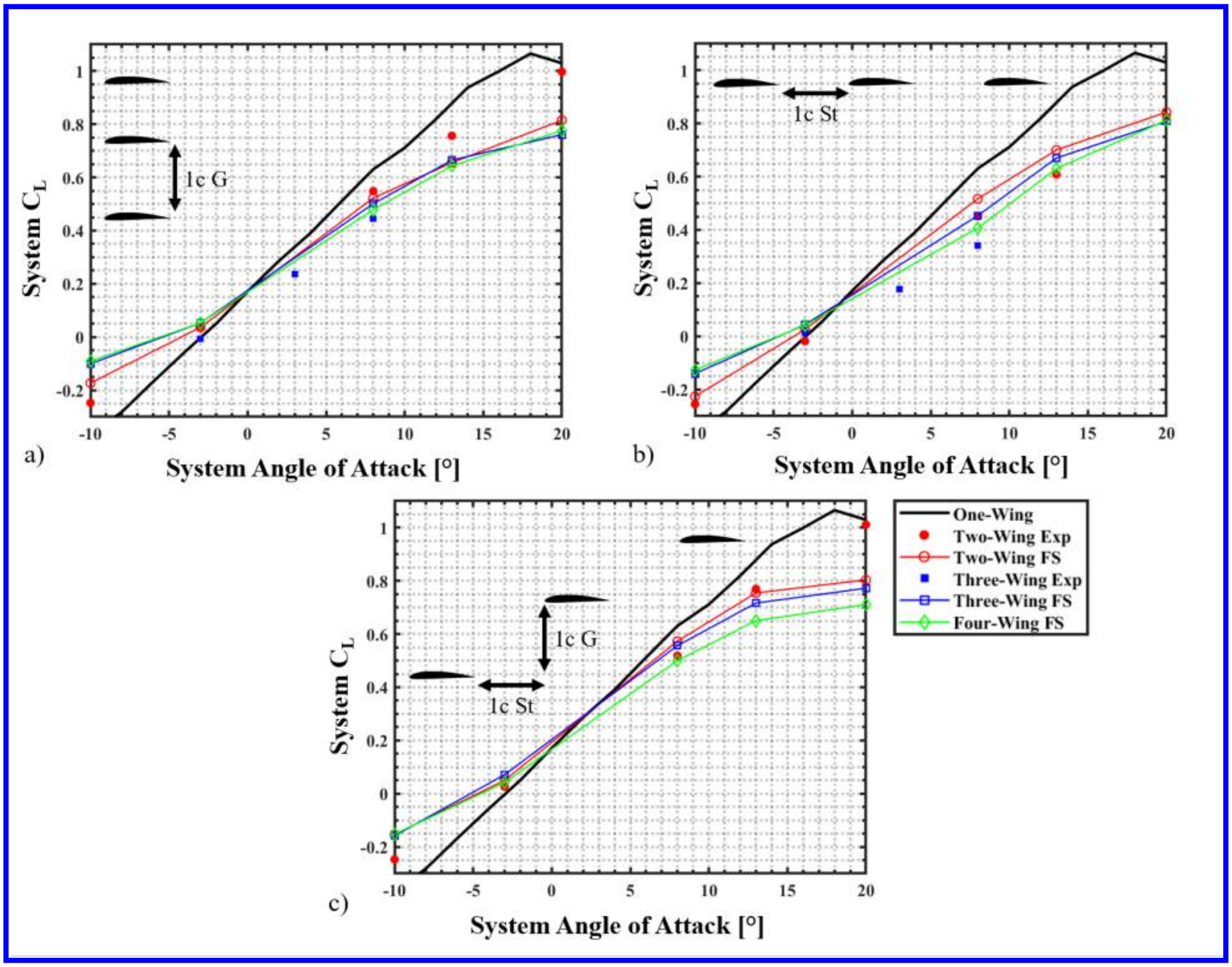

Fig. 17 Lift coefficient changes when adding wings to a) Vertically Stacked, b) Tandem, and c) Diagonal configurations

The changes in the lift curve slopes from the FlightStream results are quantified and visually represented in Fig. 18a. The lift curve slope decreases as the number of wings increases. For two wings, there is a distinct difference between the diverse configurations, but they all converge to a similar lift curve slope at four wings. The lift curve slopes of the tandem and diagonal configurations follow nearly identical trends, whereas the lift curve slope of the vertically stacked configuration remains relatively constant, regardless of the number of wings. Overall, the trend of decrease in lift curve slope with increase in number of wings aligns with the results from Mongin et al. [9]. The reduction in lift curve slope in Mongin et al. [9] could be attributed to the changes in wingspan as the number of wings was increased. However, in the present study, the span of the wings was kept constant and yet Fig. 16a shows a decrease in the lift curve slope. This suggests that the decrement could be partly due to the wing-wing interactions rather than solely AR effects. Weak senstivity to the stacked, tandem, and diagonal configurations beyond four wings says that their must be another effect which is dominating the decrease in lift curve slope as a function of the number of wings. The results presented to the right are overlaid with the findings from the previous distributed lift study by Truszkowski et al. [7] in Fig. 18b. In both the studies, the general trend shows that as the number of wings increased, the lift curve slope decreases. Results in Fig. 18b indicate that regardless of AR effects, the lift curve slope decreases when the number of wings is increased. While the configurations tested in this work did not surpass four wings, it is hypothesized that a similar plateau would be present after five wings as seen with the Truszkowski results.

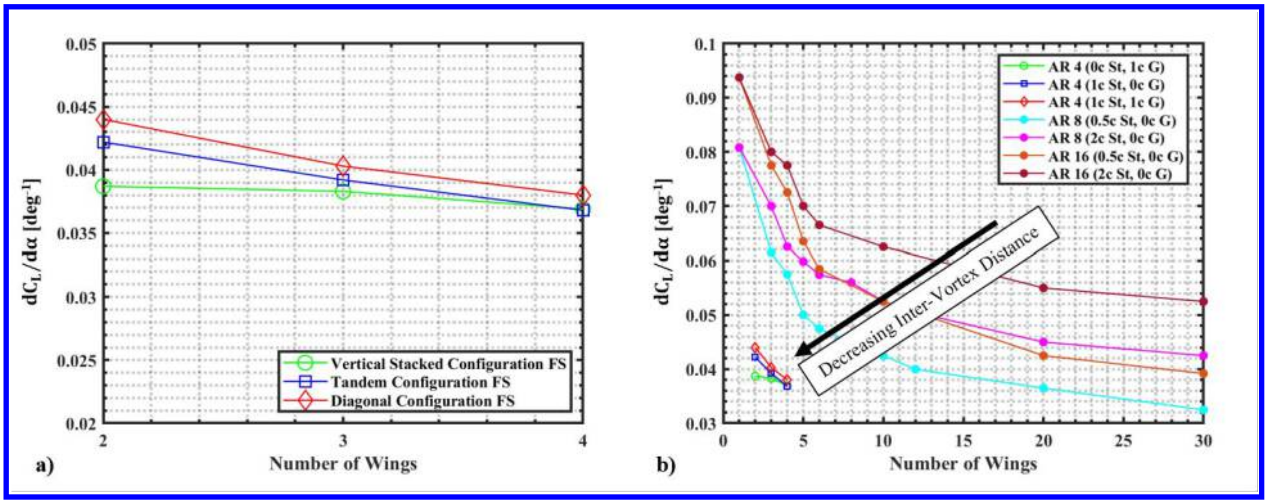

Fig. 18 Effect of adding wings on the lift curve slope from a) current study and b) compared to previous studies

From the FlightStream lift curve slopes, the effective aspect ratio can be back calculated. Two methods are used to account for different AR ranges and highlight the different assumptions. As used above in Fig. 10 of the single wing baseline, Helmbold determines the effective ARs for the various configurations with two, three, and four wings. With the semi-span AR being low, Slender-Wing Theory in Eq. (4) [19] was utilized and plotted along with the Helmbold results in Fig. 19. While the two methods differ in magnitude, both continue to emphasize the decrease in effective aspect ratio as wings are added until a plateau occurs. This same analysis was performed with similar trends by Mongin et al. [9], but this time a reduction in span was not a factor therefore providing further evidence of complex interactions.

$$C_L = \frac{1}{2}\pi AR\alpha \tag{4}$$

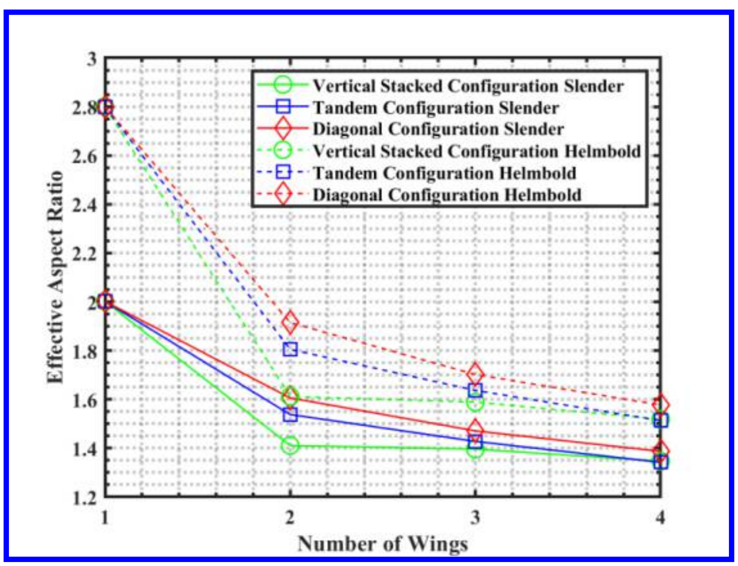

Fig. 19 Calculated effective aspect ratio as a function of number of wings and configuration

Overall, the FlightStream simulations exhibit reasonable agreement within the proximities validated, especially in the intermediate attached flow regime. To collect more information about configurations tested experimentally and validated in terms of aerodynamic forces, the coefficient of pressure (C_P) surface contours are extracted from FlightStream and plotted in two different views to observe the upper and lower surfaces. Streamlines are shown in one of the subplots to provide context regarding the general direction of the flow.

In Fig. 20a of the biplane configuration, when both wings are at an angle of attack of 8°, they individually converge on the same C_L. Theoretically, for both wings to generate the same C_L, the pressure difference between the upper and lower surfaces of the wing must be identical. However, the pressure contours on the upper surface are predicted by FlightStream differ among the wings as displayed in Fig. 20a. The top wing shows a deeper blue, indicating a lower pressure compared to the bottom wing. At this gap spacing of 1c G, it is likely that the pressure side on the top wing is affected by the low pressure on the suction side of the bottom wing. This is evident by looking at the C_P on the lower surface of the top wing which is only a small magnitude when on the bottom wing it is around a magnitude of 0.2. The addition of a fourth wing is shown on the right in Fig. 20c. While the top wing does exhibit a high magnitude low pressure region on the upper surface compared to the three wings below, it also has a low pressure region on the lower surface more than those wings below it. The trend of this pressure mixing continues to get stronger to the point that a wing can have a low pressure region on both surfaces.

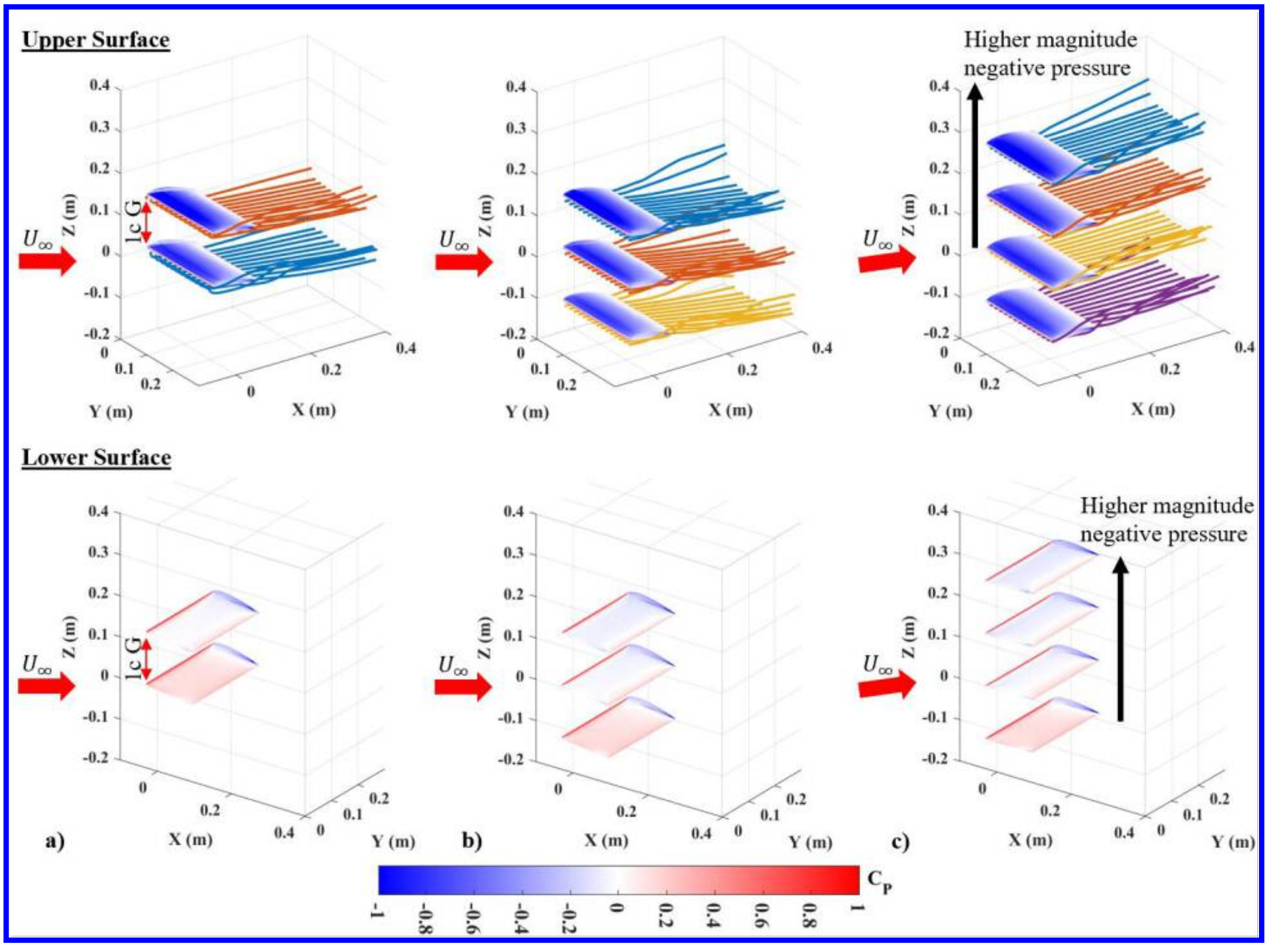

Fig. 20 C_p surface contours of both top and bottom isometric views for a) two wings, b) three wings, and c) four wings in a vertical stack

For three wings in tandem set at an angle of attack of 8°, each wing generates a different lift force. In both experiments and FlightStream, the general trend observed is a progressive decrease in lift generation from the front wing to the aft wing. The streamlines depicted in Fig. 21a illustrate how the streamlines from each wing intermingle and expand as the wakes extend downstream. As the wakes become more dominant, they exert a strong effect on the lower surfaces of the downstream wings, leading to a decrease in their high pressure. This, in turn, reduces the overall pressure difference across the wings. Figure 21b adds a fourth wing to the tandem configuration. As expected, the rearmost wing experiences the smallest pressure difference with a lower magnitude of low pressure on the top and only a minuscule area of high pressure on the trailing edge. While not shown in the figures, the fourth wing's surface is heavily influenced by the expansion of the wingtip vortices generated by the wings upstream.

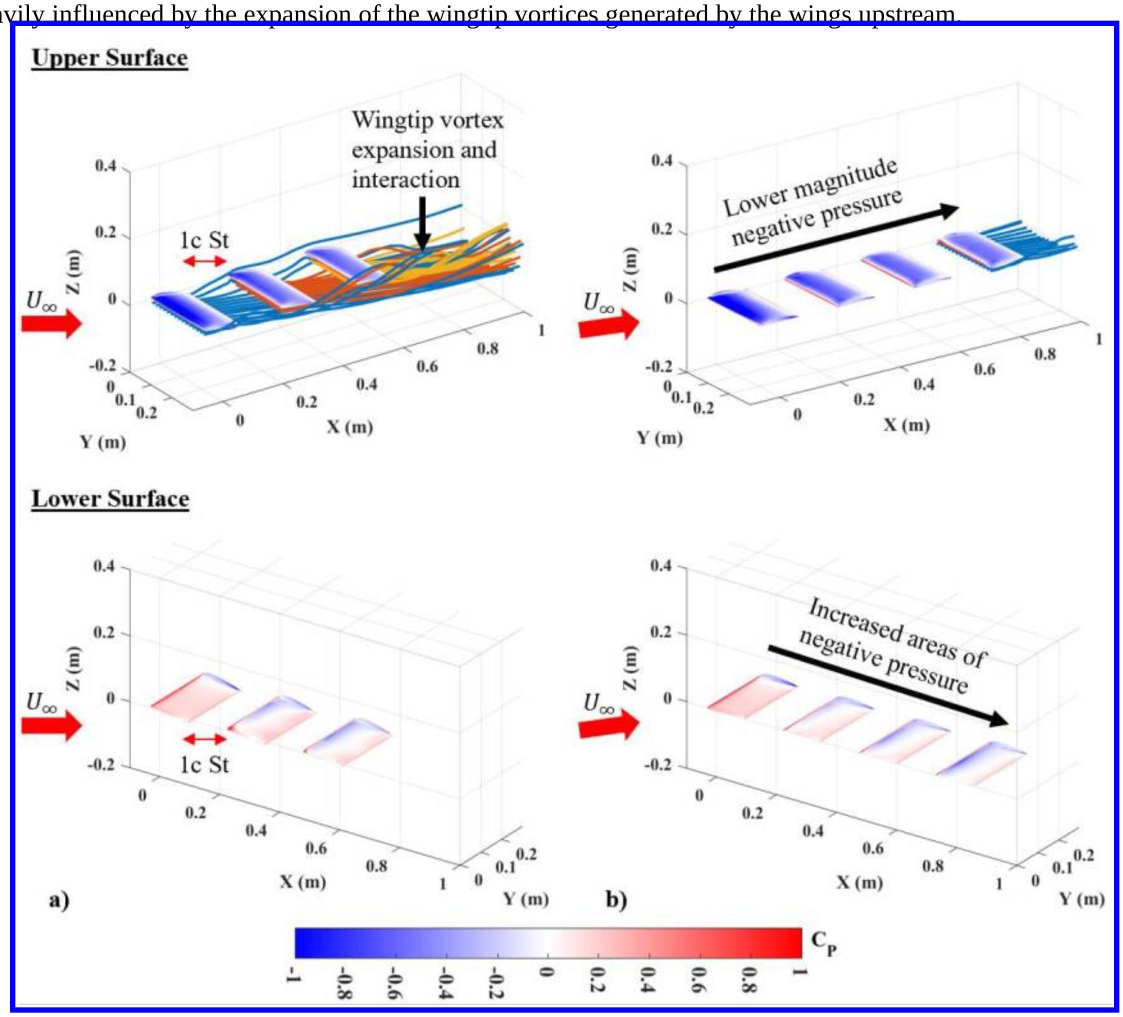

Fig. 21 C_P surface contours of both top and bottom isometric views for a) three wings and b) four wings in tandem

Another consequence of adding wings to a base configuration is that the lift curve slope of the wings shifts to the right, mimicking an increased camber effect while maintaining the lift curve slope. Therefore, it is hypothesized that the shift is caused by the changes in the induced angle of attack due to the downwash. In order to quantify these effects, individual lift coefficient curves for each wing will be studied for a vertical stack and tandem configuration with different numbers of wings. Two general methods are used to determine the difference in induced angle of attack for each wing. For a Sweeping Wing, the measured $\alpha_{L=0}$ subtracted by the true baseline Clark Y $\alpha_{L=0}$ of 3° is the $\Delta\alpha_i$. For the fixed angle of attack Neighbor Wings, an approximation is used to determine the Single-Wing Baseline angle of attack that achieves roughly the same average measured $C_{L_{Neighbor}}$.

Figure 22 presents a comparison of the induced angle of attack effects on individual wings in either a two-wing vertical stack or three-wing vertical stack where the Sweeping Wing is on the bottom and the Neighbor Wing(s) are fixed at 8°. Following the process explained above, the Sweeping Wing in the two wing configuration experiences it's $\alpha_{L=0}$ at -1° therefore has a $\Delta\alpha_i$ of 2°. The Neighbor Wing is trickier as it experiences a large fluctuation through the Sweeping Wing sweep. When a third wing is added above, the Sweeping Wing has a larger $\Delta\alpha_i$ of 3° by having $\alpha_{L=0}$ at 0°. The middle Neighbor Wing has a large fluctuation just as observed with the two-wing configuration making it difficult to quantify the $\Delta\alpha_i$, but the top Neighbor Wing has around a $\Delta\alpha_i$ of 4° with it achieving a lift coefficient that is similar to a baseline Clark Y at 4°.

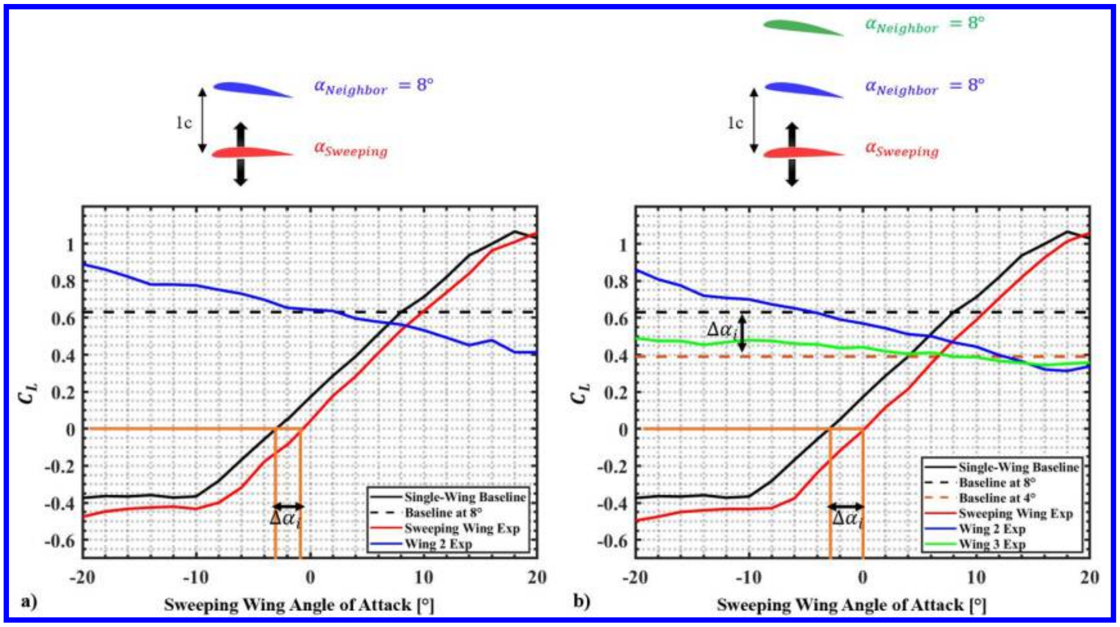

Fig. 22 Induced angle of attack effects on a) two wings and b) three wings in a vertical stack

Figure 23a plots a two-wing tandem configuration where the Sweeping Wing is placed behind the front Neighbor Wing fixed at 8°. As seen by the minimal difference between the blue line of the front wing and the dashed black line representing the lift coefficient of the Single-Wing Baseline at 8°, the Neighbor Wing experiences relatively feeble proximity effects. The Sweeping Wing on the other hand experiences an $\alpha_{L=0}$ around 3°, not at the -3° of the standalone wing. The effects of downwash from the front wing are assumed to be responsible for the $\Delta\alpha_i$ of 6° of the rear wing as it sweeps. Figure 23b now plots a three-wing tandem configuration where a second fixed angle of attack wing is added upstream of the Sweeping Wing as shown. The front wing performs closely to the expected standalone wing at 8°, but the wings downstream show an offset from their expected performance. The middle wing while fixed at 8° actually generates the lift corresponding the Single-Wing Baseline at 2° showing that the $\Delta\alpha_i$ is 6°, which is consistent with what the aft wing experienced in the two-wing configuration. The aftmost wing, the Sweeping Wing, achieves $\alpha_{L=0}$ at 5° resulting in an adjusted $\Delta\alpha_i$ of roughly 8°.

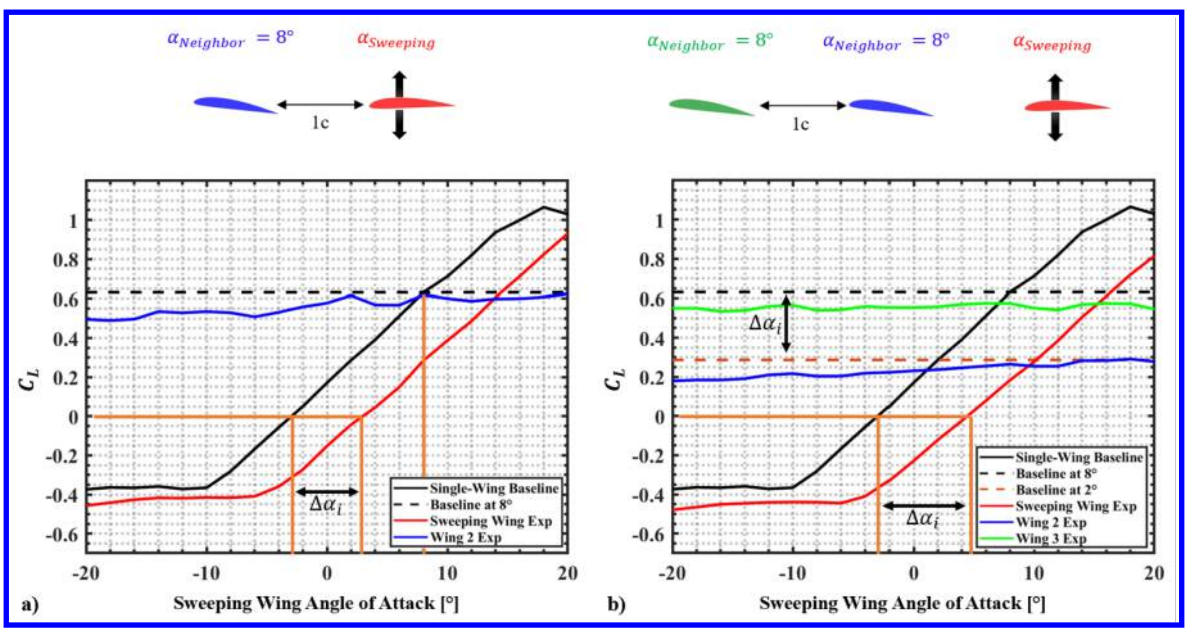

Fig. 23 Induced angle of attack effects on a) two wings and b) three wings in tandem

Preliminary work has been undertaken to explore downwash adjustment, whereby the rear wings are set at a higher angle of attack to counter any reduction in the induced angle of attack. The study is aimed at answering the question of whether a multi-wing layout with significant décalage can achieve the mono-wing efficiency that has so far been elusive in distributed lift studies focused on proximity. The optimization of a three-wing tandem configuration, with 1c St between the wings, was attempted using this methodology. The front wing (Wing One) was placed at 4°, which is roughly around the maximum L/D of a standalone Clark Y wing. A second wing was added and swept until it hit its own maximum L/D while the front wing was locked at 4°. Finally, a third wing was added and swept in a similar process with the forward two wings at their maximum L/D. The end result is presented in a bar chart form in Fig. 24. Basic flow visualization, in Fig. 25 was conducted using string trailing edge tracers on all three wings to understand the direction of the downwash of a given configuration to confirm if direct impingement was occurring.

As witnessed in Fig. 23, upwash effects on performance from the downstream wing back to the upstream wing seem limited. The foremost wing performs similarly to the standalone wing. The wing with one wing upstream experiences a $\Delta\alpha_i$ of 6° whether or not there is another wing downstream. This is manifested as minute changes on the bar graph as experienced on the first and second upstream wings through the two and three wing configuration. It is established that even at high angles of attack [adjusted for effective induced angles], the downstream wings are unable to achieve a similar maximum L/D as their companion wings upstream. When extracting the system L/D for the two-wing and three-wing configurations, it is found that the system L/D is significantly larger than those presented in Fig. 13 for configurations without this induced angle of attack adjustment. The two-wing configuration achieved a system L/D of 15.3 and the three-wing had a system L/D of 17.7 both of which are higher than any value in Fig. 15. The flow visualization showed that when arranged in this décalage combination, the downwash direction does not seem to impact the surface of the downstream wings but passes below the lower surfaces in a parallel direction. This could be responsible for why these angles of attack produce the highest L/D values.

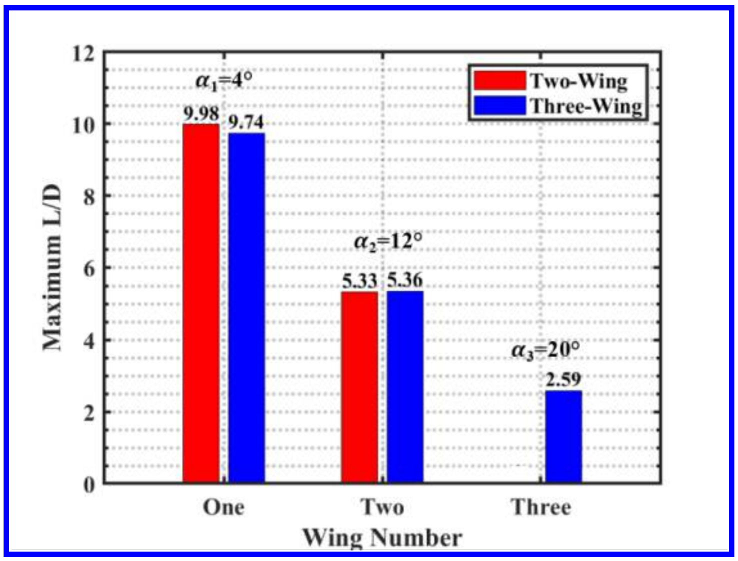

Fig. 24 Maximum individual wing L/D for two-wing and three-wing tandem configuration

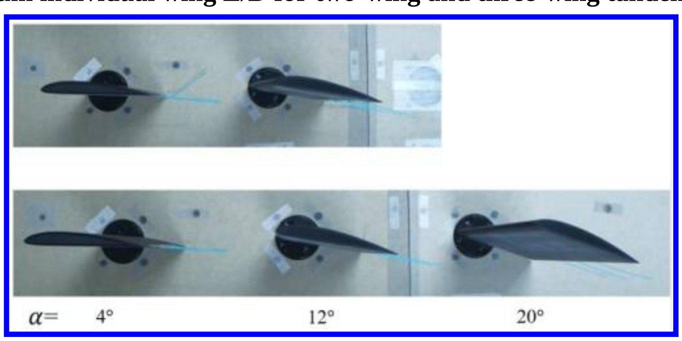

Fig. 25 Trailing Edge Tracer flow visualization with cyan string on two-wing (top) and three-wing (bottom) tandem configuration

V.Conclusions

Several different configurations of two, three, and four Clark Y AR 2 semispan wings at various staggers, gaps, and décalage angles were experimentally and numerically investigated. The data collected was analyzed through local individual wing performance and global system performance. The results quantified the wing-wing interactions as a function of stagger, gap, décalage, and number of wings.

Adding wings proved universally decremental in performance for the system lift curve slope and maximum system L/D. For four-wing configurations, mutually negative wing-wing interactions were dominant irrespective of the position of a sweeping or a fixed wing. No configuration tested within the experimental proximity ranges for the present study differed significantly from one another. None of the configurations were able to achieve more than 60% of the Single-Wing maximum L/D. For multi-wing designs, this study has shown that within the parameter space explored in this study, adding wings beyond a certain point will not achieve the total aerodynamic performance of a single wing. Hence, selection of a distributed lift configuration would likely be influenced more by a design constraint driven by one of the earlier stated advantages of the configuration.

More specifically, the addition of wings on certain configurations such as a vertical stack, tandem, and diagonal is evaluated using aerodynamic force measurements and surface pressure contours. Based on experimental results, system lift curve slope decreased as wings were added irrespective of the wing arrangements. When arranged in a vertical stack, there was slightly less sensitivity to the addition of wings relative to the tandem and diagonal configurations. At four-wings, the previously mentioned convergence of performance was observed once again for these three extreme configurations. Numerical simulations showed the possible presence of some surface pressure interactions of upper and lower surfaces when in the vertical stack and possible reduction in surface pressure due to wingtip vortices in the tandem. Plotting the individual lift coefficients of wings in the above base configurations shows changes in the induced angle of attack on each wing. When in a vertical stack, this induced angle effect is minimized, but in a tandem configuration it is magnified with large changes in induced angle of attack up to a decrease of 8° on the third wing downstream.

Adjusting the downstream wings to a higher angle of attack showed an inability to match the efficiencies of the upstream wings but did result in a higher total system L/D than experienced before. Overall, while basic trends were established that can help guide multi-wing placements, more exploration must be done to understand if more system and individual wing efficiency can be achieved.

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